

Computer Networks and TCP

Introduction

Computer networks allow devices to communicate and exchange data. A network may contain computers, routers, switches, servers, and other devices.

TCP

Transmission Control Protocol (TCP) is a transport-layer protocol that provides reliable, connection-oriented communication. TCP uses sequence numbers, acknowledgements, retransmission, and flow control to deliver data reliably.

IP

Internet Protocol (IP) is responsible for addressing and routing packets between networks. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses.

DNS

The Domain Name System (DNS) translates human-readable domain names into IP addresses. DNS uses records such as A, AAAA, CNAME, and MX records.

Test Keywords

This document intentionally contains the terms: TCP, reliable communication, sequence numbers, acknowledgement, IP address, routing, DNS, domain name, and network.